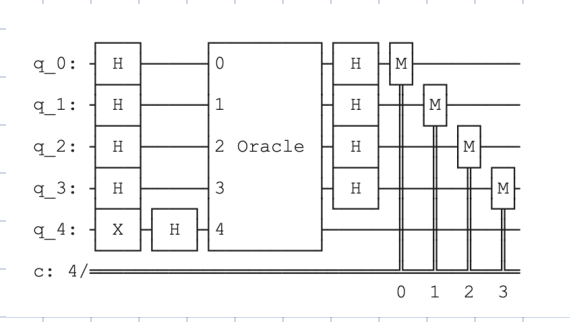
DJ Circuit from last lecture



Grover's algorithm

rather: find x, s.t $f(x) = 1$

- algorithm "searching an unsorted database" with $N = 2^n$ elements.

  in $O(\sqrt{N})$ elements. either: find x, s.t $f(x) = 1$.

→ classical algorithm $\frac{N}{2} = O(N)$ time.

goal: find w, given an Oracle U_f with $f: \{0,1\}^n \to \{0,1\}$.

$$f(x) = \begin{cases} 1, \text{ if } x = w \\ 0 \text{ else} \end{cases}, \quad f_0(x) = \begin{cases} 0, \text{ if } x = 00\ldots0 \\ 1 \text{ else} \end{cases}$$

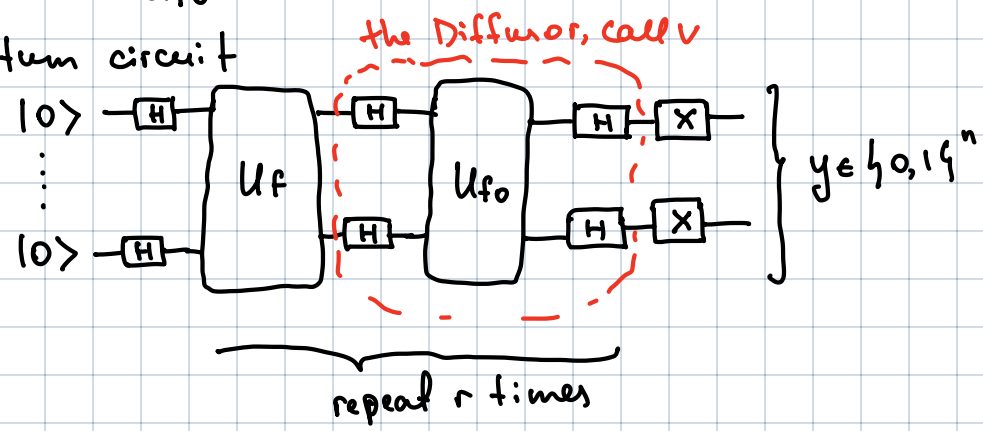
$$\underbrace{U_f |x\rangle = (-1)^{f(x)} |x\rangle}_{\text{remember from before.}} \quad \hookrightarrow \quad U_f : |w\rangle \to -|w\rangle$$
$$|x\rangle \to |x\rangle \quad \forall x \neq w.$$

$$U_f = I - 2|w\rangle\langle w|$$

$$\Rightarrow U_{f_0}: \; |0\rangle^{\otimes n} \to |0\rangle^{\otimes n}$$
$$|x\rangle \to -|x\rangle \quad \forall x \neq 00\ldots0$$

$$U_{f_0} = 2|0\rangle\langle 0|^{\otimes n} - I.$$

- quantum circuit



Claim: $y = w$ (with high probability)

Proof: Let us define uniform superposition state.

$$|s\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle$$

and

$$V = H^{\otimes n} U_{f_0} H^{\otimes n} = H^{\otimes n}\cdot 2|0\rangle\langle 0|^{\otimes n} H^{\otimes n} - H^{\otimes n}\cdot I\cdot H^{\otimes n} =$$

$$= 2\cdot \underbrace{H^{\otimes n}|0\rangle^{\otimes n}}_{|s\rangle}\underbrace{\langle 0|^{\otimes n}H^{\otimes n}}_{\langle s|} - I = 2|s\rangle\langle s| - I$$

$|0\rangle - \boxed{A} - \boxed{B} -$

$B\cdot A\cdot|0\rangle$

"approach from the left"

$\Rightarrow$ Grovers algorithm carries out the operation $(V\cdot U_f)^r$ on the state $|s\rangle$.

all possible linear comb. of these vectors.

Let Σ be the plane spanned by $|s\rangle$ and $|w\rangle$.

and let $|w^\perp\rangle$ be the state orthogonal to $|w\rangle$ in Σ:

So $|w^\perp\rangle := \frac{1}{\sqrt{2^n-1}}\sum_{\substack{x\in\{0,1\}^n\\ x\neq w}}|x\rangle \Rightarrow \sum_{\substack{x\in\{0,1\}^n\\ x\neq w}}|x\rangle = \sqrt{2^n-1}\,|w^\perp\rangle$

So: $|s\rangle = \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle = \frac{1}{\sqrt{2^n}}\left(|w\rangle + \sum_{\substack{x\in\{0,1\}^n\\ x\neq w}}|x\rangle\right) =$

$$= \frac{1}{\sqrt{2^n}}\left(|w\rangle + \sqrt{2^n-1}\,|w^\perp\rangle\right) = \frac{|w\rangle}{\sqrt{2^n}} + \sqrt{\frac{2^n-1}{2^n}}\,|w^\perp\rangle =$$

$$= \sin\frac{\theta}{2}|w\rangle + \cos\frac{\theta}{2}|w^\perp\rangle$$

define θ, s.t

$$\sin\frac{\theta}{2} = \frac{1}{\sqrt{2^n}}$$

$$\Downarrow$$

$$\theta = 2\cdot\arcsin\frac{1}{\sqrt{2^n}}$$

$$\sin^2\frac{\theta}{2} + \cos^2\frac{\theta}{2} = 1$$

$$\Rightarrow \frac{1}{2^n} + \cos^2\frac{\theta}{2} = 1 \Rightarrow$$

$$\Rightarrow \cos^2\frac{\theta}{2} = \frac{2^n-1}{2^n}$$

$$\Rightarrow \cos\frac{\theta}{2} = \sqrt{\frac{2^n-1}{2^n}}$$ check out.

what if we plot this

<u>protocol</u>: 1. Prepare $|s\rangle$

2. Apply $U_f = I - 2|w\rangle\langle w|$

reflection at $|w^\perp\rangle$

3. Apply $V = 2|s\rangle\langle s| - I$

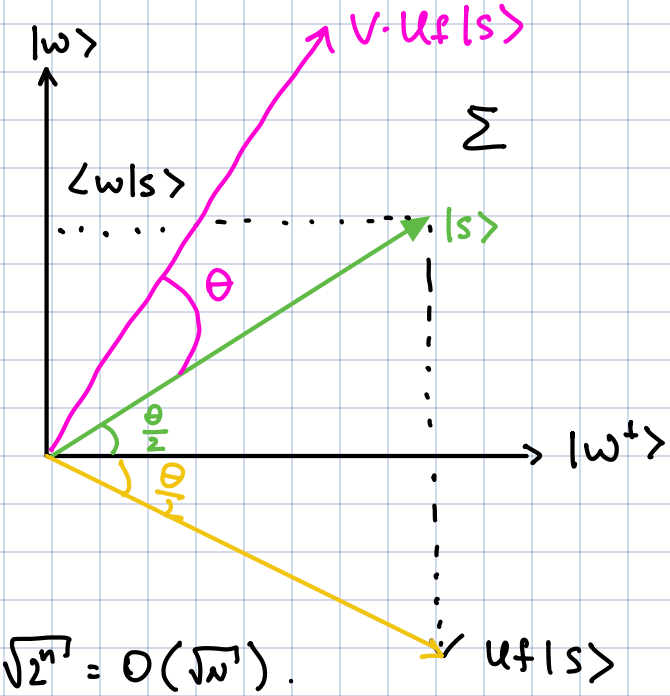
↳ reflection at $|s\rangle$

⇒ $V \cdot U_f$ correspond to a rotation by an angle θ

⇒ after r applications of 2.) & 3.), the state is rotated by $r \cdot \theta$.

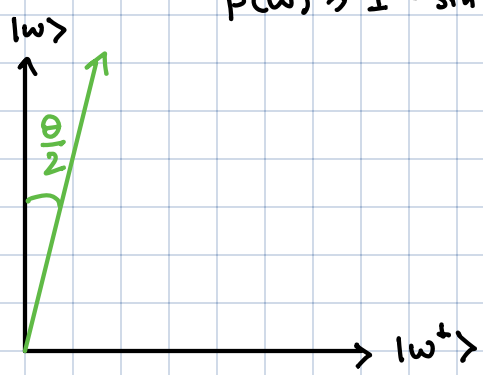
↳ choose r, s.t. $r \cdot \theta + \frac{\theta}{2} \overset{!}{\approx} \frac{\pi}{2}$

↳ $r = \frac{\pi}{2\theta} - \frac{1}{2}$.



So: $r = \frac{\pi}{4 \cdot \arcsin\left(\frac{1}{\sqrt{2^n}}\right)} - \frac{1}{2} \approx \frac{\pi}{4} \cdot \sqrt{2^n} = O(\sqrt{N})$.

n is large.

⇒ after r calls to the oracle, the final measurement will result is state $|w\rangle$ with min prob.

$p(w) \geq 1 - \sin^2 \frac{\theta}{2} = 1 - \frac{1}{2^n}$ ← if n is high we would be close to the answer.